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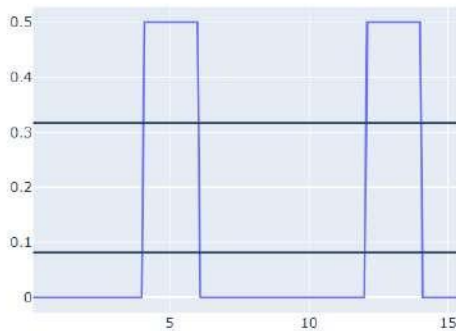
Course: ECA3102

Nano electronics

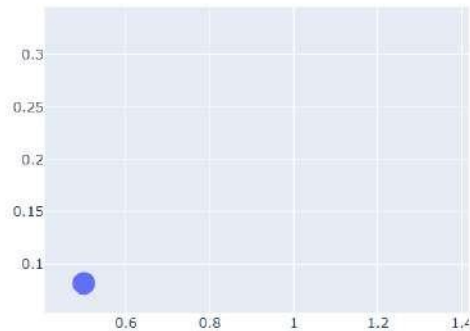
Experiment 5: Demonstration of Quantum Tunneling

Aim: To demonstrate the concept of **quantum tunneling**, where electrons or other quantum particles pass through a potential energy barrier that cannot be crossed according to classical physics, and to study its applications in modern electronic devices.

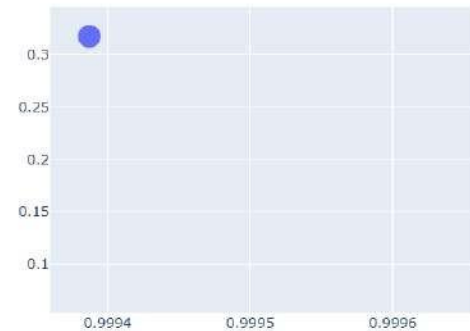
Potential Energy vs Distance



Resonance Peaks vs Occurrence



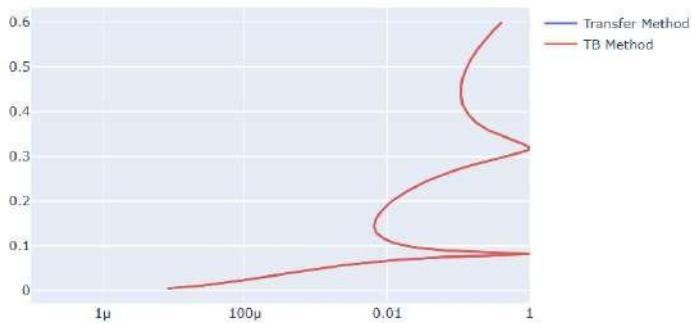
Transmission Peaks vs Energy



Aim: To analyze the variation of the transmission coefficient (T) with respect to the incident particle energy (E) using a refined computational grid, and to study the quantum tunneling behavior through a potential barrier with improved numerical accuracy.

- 1) At low energies, the transmission coefficient is very small.
- 2) As the particle energy increases, the transmission coefficient gradually increases.
- 3) Near and above the barrier energy, the transmission coefficient approaches 1 (100% transmission).

Transmission Coefficient Vs Energy on refined grid



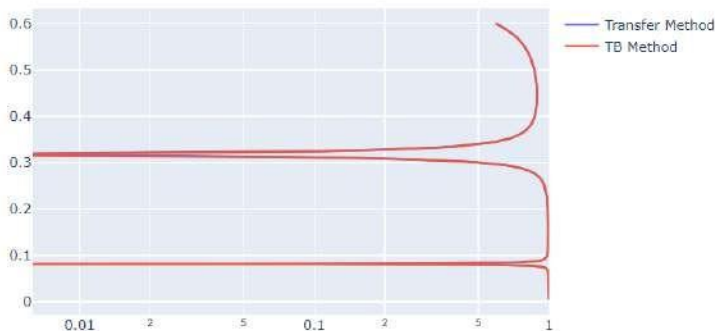
Case study 2: Analysis of Reflection Co-efficient vs Energy on a Homogeneous Grid

Aim: To analyze the variation of the **reflection coefficient (R)** with respect to the **incident particle energy (E)** using a **homogeneous (uniform) computational grid**, and to study the reflection characteristics of quantum particles interacting with a potential barrier.

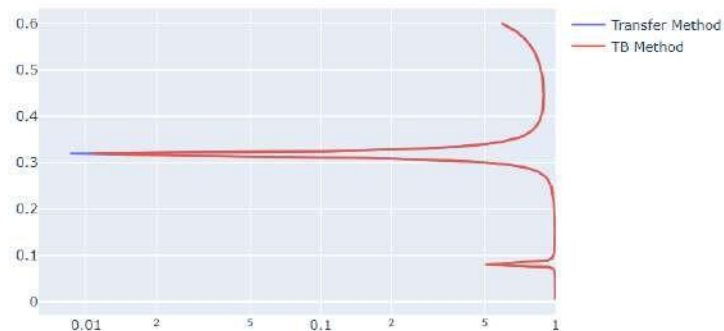
Observation:

- 1) At **low energies**, the reflection coefficient is close to **1 (complete reflection)**.
- 2) As the particle energy increases, the reflection coefficient decreases.
- 3) At energies higher than the barrier height, the reflection coefficient approaches **0 (negligible reflection)**.

Reflection Coefficient Vs Energy on refined grid



Reflection Coefficient Vs Energy on Homogeneous Grid



Conclusion: The two case studies demonstrate the fundamental principle of quantum tunneling by analyzing the transmission and reflection behavior of particles encountering a potential barrier.

- Case Study 1 shows that the transmission coefficient increases with increasing particle energy, indicating a higher probability of tunneling through the barrier.
- Case Study 2 shows that the reflection coefficient decreases as particle energy increases, meaning fewer particles are reflected when they possess higher energy.
- Since Transmission Co-efficient (T) + Reflection Co-efficient (R) = 1, the results satisfy the law of probability conservation in quantum mechanics.
- Overall, these analyses validate the quantum mechanical prediction that particles can tunnel through a barrier even when their energy is lower than the barrier height, a phenomenon that has important applications in tunnel diodes, scanning tunneling microscopes (STM), flash memory, quantum devices, and semiconductor technology.